



Predicting *E. coli* Drug Resistance

*Using Traditional Machine Learning
and Deep Learning*

Abdoulfatah Mohamed Abdillahi

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COMMITTEE

Dr. Sara El Alaoui · Dr. Akila de Silva · Dr. Pleuni S. Pennings

How this talk is organized.

01

Background & Motivation

Antimicrobial resistance in E. coli and why pangenome-based approaches matter for AMR prediction.

02

Methods

Pangenome construction from 588 isolates, gene categorization, and dual feature extraction (SNP & GPA matrices).

03

Results

Model comparison across RF, LR, XGB, and TabR; biological validation of gyrA and parC resistance signal.

04

Conclusion & Future Work

Key findings, limitations, and directions for unified SNP+GPA representations and deep learning extensions.

Antibiotic Resistance: A Global Health Crisis

>1M¹

deaths annually
from antimicrobial resistance

¹ Murray et al., *The Lancet*, 2022

#1

E. coli — most common
Gram-negative pathogen

Days

to get results from
culture-based testing

Whole-genome sequencing combined with machine learning offers a faster and more scalable approach.

Why Ciprofloxacin in *E. coli*?

Clinical Relevance

- *E. coli* is the most common cause of urinary tract and bloodstream infections.
- Ciprofloxacin is widely used to treat severe *E. coli* infections.
- However, resistance to ciprofloxacin is increasing globally.

¹ Kot, *Polish J. Microbiology*, 2019

From Biology to Feature Design

Chromosomal Mutations → SNP-based Features

- Mutations in key genes (e.g., *gyrA*, *parC*) drive resistance
- Captured as SNPs in the model.

Horizontal Gene Acquisition → GPA Features

- Bacteria can acquire resistance genes from other organisms
- Represented as gene presence/absence features

→ This dual mechanism provides a natural framework for comparing SNP-based and GPA feature representations.

Where Does the Data Come From?

All isolates from human bloodstream infections (bacteraemia) — clinically consistent across all three studies.

200 isolates¹

Invasive E. coli — England

¹ Moradigaravand et al., *PLOS Comp. Biol.*, 2018

EUCAST breakpoints applied

200 isolates²

NORM Surveillance — Norway

² Gladstone et al., *The Lancet Microbe*, 2021

EUCAST breakpoints applied

200 isolates³

US Military Healthcare Network

³ Mills et al., *Genome Medicine*, 2022

EUCAST breakpoints applied

600

initial isolates



18

intermediate
reclassified → susceptible (EUCAST)



-12

excluded
(no genomic data)



588

final matched isolates

Two ways to read a genome.

Two complementary genomic signals used for prediction.

SNPs · Single Nucleotide Polymorphisms

Letter-level differences in DNA at the same position.

Isolate 1	A	T	G	C	A	T
Isolate 2	A	T	G	T	A	T
Isolate 3	A	T	G	C	A	T

↑ SNP

- Captures point mutations (e.g., *gyrA S83L*)
- Same gene across strains — sequence varies
- Output: **SNP matrix** — $600 \times 166,415$ features

GPA · Gene Presence / Absence

Map gene to isolate → encode by coverage.

	Alignment of <i>blaTEM</i> gene											Encoded
Isolate 1	A	T	G	C	A	T	G	C	T	A	100%	→ 1
Isolate 2	—	—	—	—	—	—	A	—	—	—	10%	→ 0
Isolate 3	A	T	G	C	A	T	G	C	T	A	100%	→ 1

Rule: coverage $\geq 50\%$ → **1 (present)** · coverage $< 50\%$ → **0 (absent)**

- Captures acquired resistance genes (e.g., *blaTEM*)
- Whole genes — present in some strains, absent in others
- Output: **GPA matrix** — 600×6092 features

SNP = sequence variation | **GPA** = gene content variation

Research Aims & Hypothesis

01

Construct Features

- SNP and gene presence/absence (GPA)

02

Benchmark Models

- 5 ML models + TabR (deep learning)
- Evaluated on both SNP and GPA datasets

03

Evaluate Impact

- Feature representation vs. model choice
- Predictive performance comparison

HYPOTHESIS

SNP features capture mutation-driven resistance (gyrA, parC) and are expected to perform strongly.

GPA features capture horizontally acquired resistance and provide complementary interpretability.

PART 01

Methods

From bacterial isolates to model-ready features

Class Distribution of Final Dataset

Label Processing & Sample Matching

01 Raw labels

282 R · 300 S · 18 I

02 Reclassify

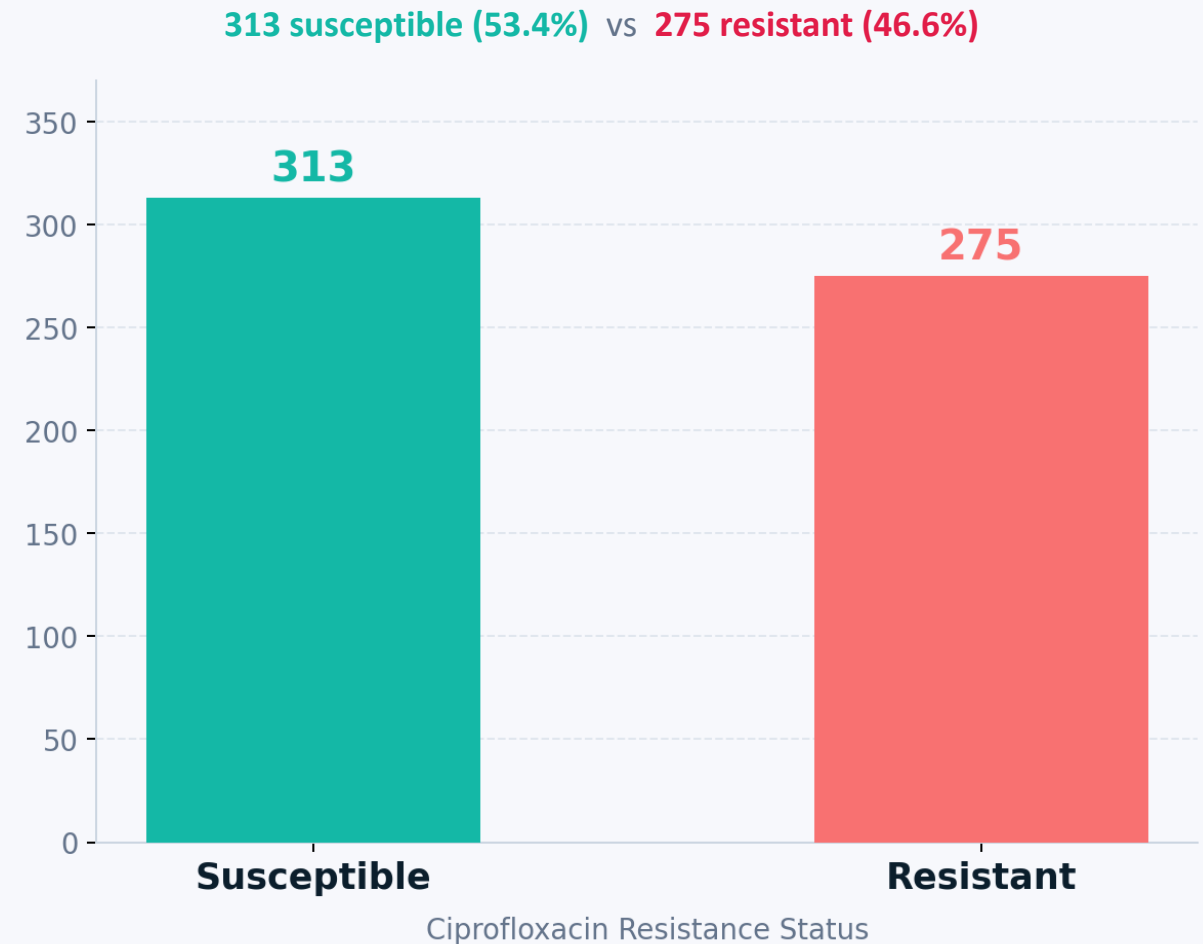
I → S (EUCAST guideline)

03 Filter

12 removed (no genomic data)

04 Final dataset

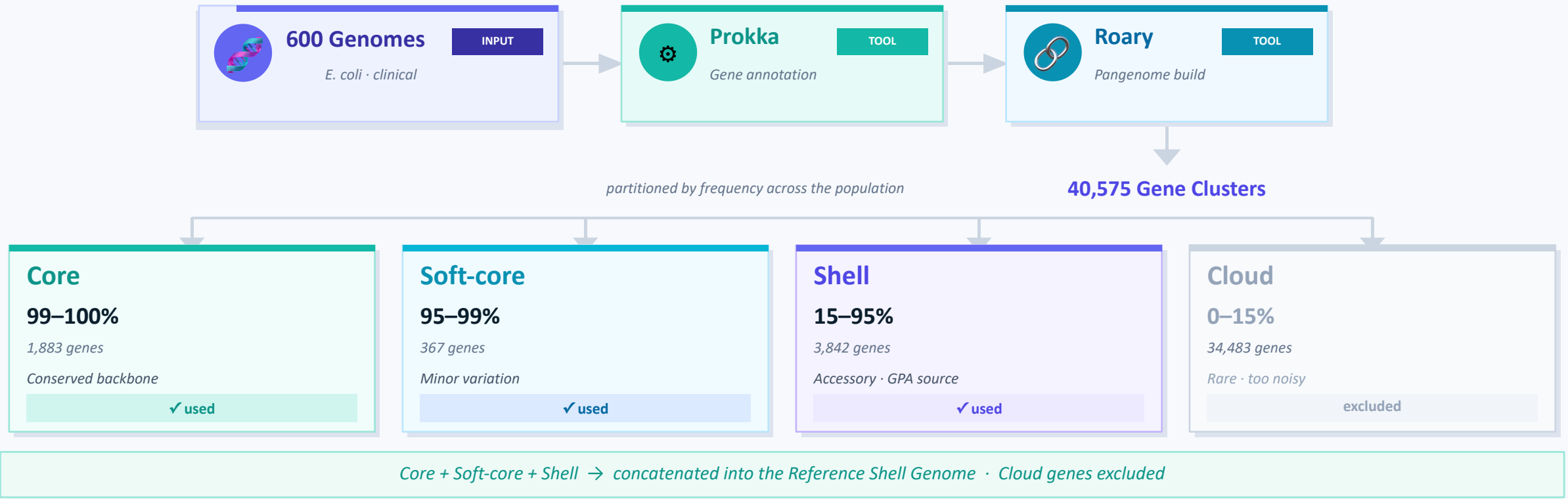
588 isolates · 313 S · 275 R



Mild class imbalance (53.4% vs 46.6%) → motivates use of Balanced Accuracy as the primary evaluation metric.

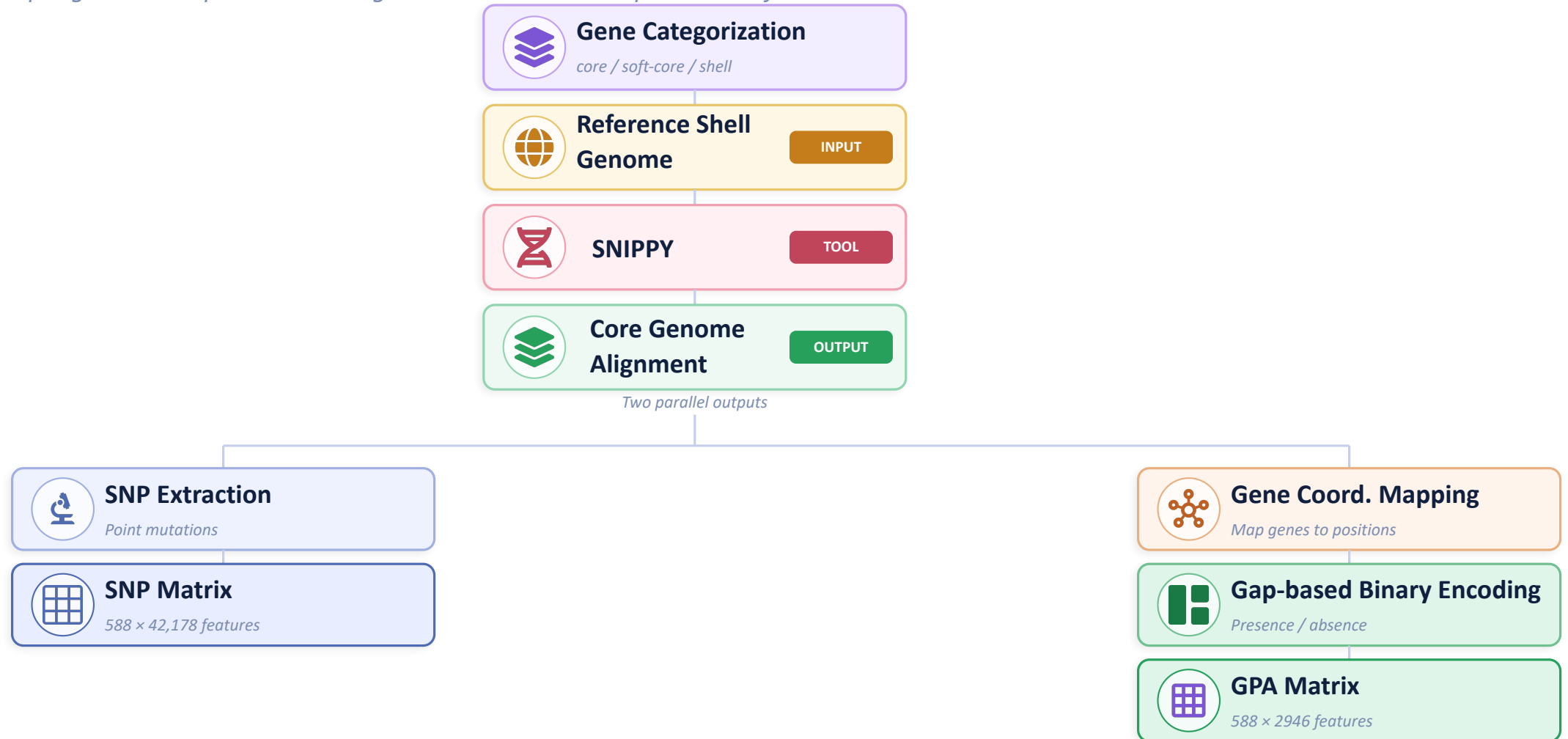
How We Built the Pangenome.

600 *E. coli* genomes annotated with Prokka · clustered with Roary · genes partitioned by frequency.



From Reference Genome to Feature Matrices.

Continuing from pangenome output · SNIPPY aligns 600 isolates → two parallel ML feature sets extracted.



SNP matrix captures mutation-driven resistance · GPA matrix captures gene-level presence/absence — two complementary views of the genome.

SNP matrix: filtering & encoding.

High-dimensional SNP data requires filtering, encoding, and feature selection before ML.

166,415 raw variant sites from core genome alignment



56,717 $\text{MAF} \geq 0.05$ & $\text{gap} \leq 10\%$ — remove rare variants & missing data



67,259 one-hot encoding of multi-allelic loci — columns expand per allele



42,178 MI feature selection (median threshold) — informative SNP features retained

⚠ *MultiSURF was computationally infeasible at 67,259 features — MI filter applied to SNPs instead.*

From Alignment to Gene Presence/Absence

Gene coordinate mapping

$$start_i = \sum_{j=1}^{i-1} length_j \quad end_i = start_i + length_i$$

with $start_1 = 0$ · genes indexed $i = 1, 2, \dots, n$

Example: cumulative positions in core alignment



$$start(atpE) = length(rpmG) = 168$$

Gap proportion

$$gap_proportion_{ij} = \frac{\text{number of gap characters in gene } i \text{ for isolate } j}{L_i}$$

Binary encoding

$$presence_{ij} = 1 \text{ if } gap_proportion_{ij} < 0.5, \text{ else } 0$$

Feature reduction

6,092

gene clusters



3,625

after constants & missingness filter



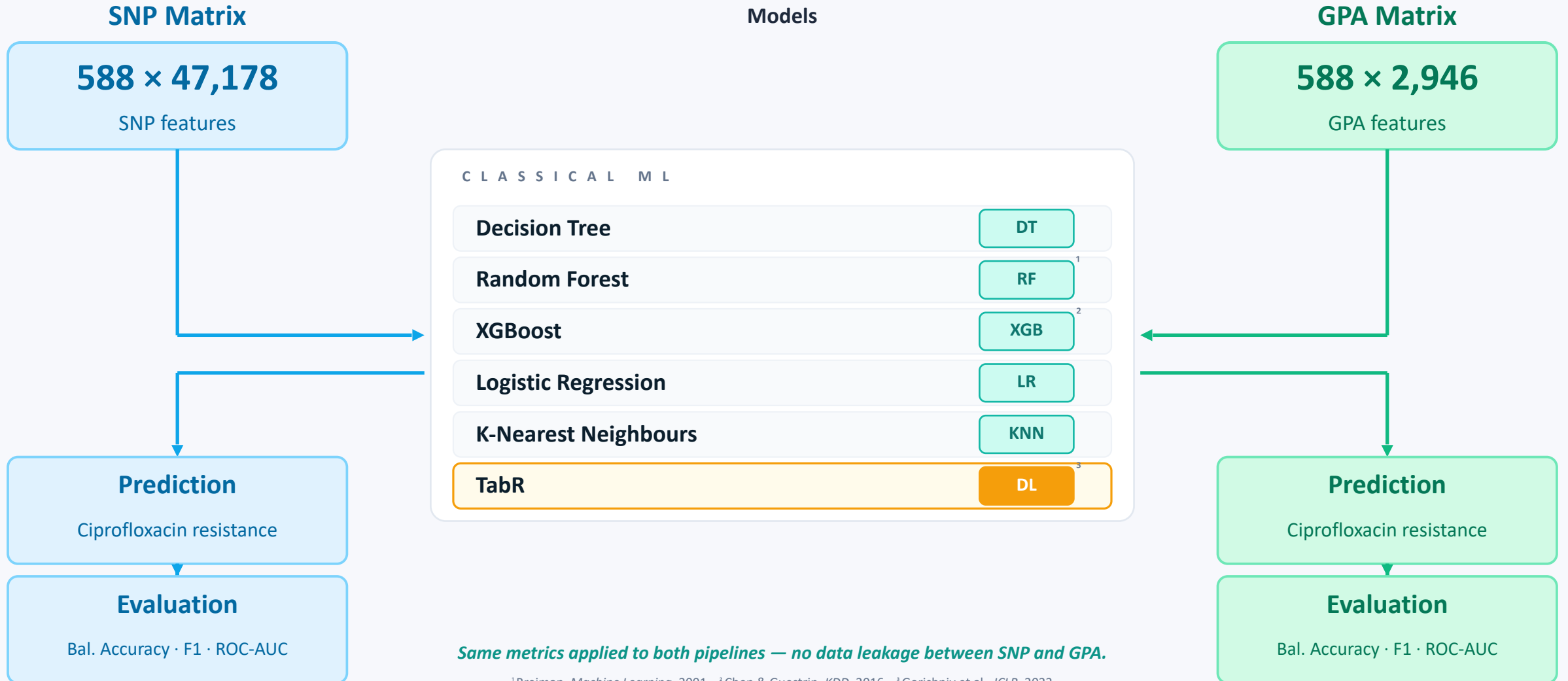
2,946

after MI + MultiSURF (median threshold)

MI captures marginal dependence · MultiSURF detects neighbourhood interactions

Model comparison across SNP and GPA features.

Same models applied independently to both feature sets — a fair, controlled comparison.

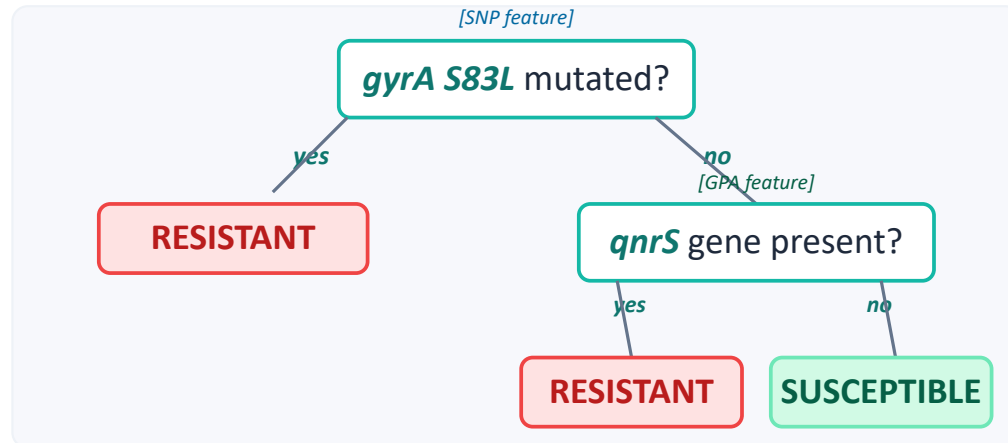


¹ Breiman, *Machine Learning*, 2001 ² Chen & Guestrin, *KDD*, 2016 ³ Gorishniy et al., *ICLR*, 2023

How the models learn.

Classical ML · learns rules

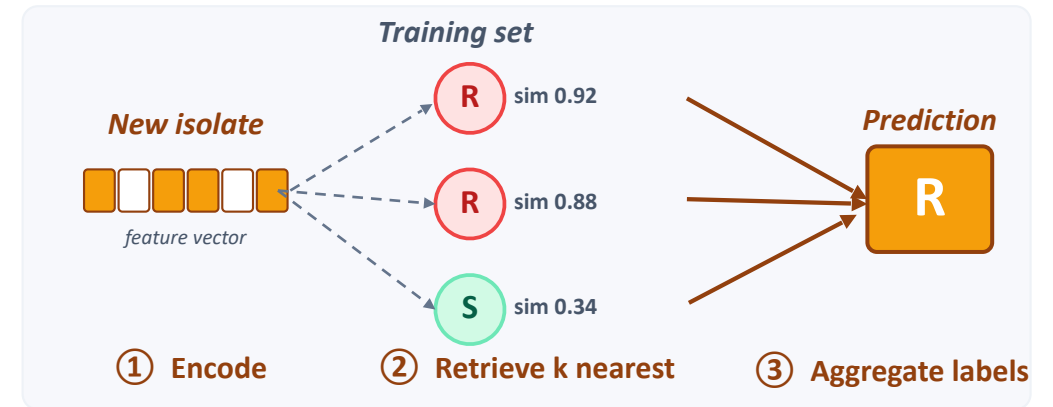
"If *gyrA* mutated → resistant. Otherwise check if *qnrS* gene is present."



- **Single concatenated matrix:** SNPs + GPA features per isolate.
- **RF / XGBoost** — many trees vote together.
- **Logistic Regression / kNN** — different decision boundaries, same input.

TabR · learns by comparison

"This isolate looks like these — predict accordingly."



- **Retrieval + aggregation** — weights neighbours by learned similarity.
- No hand-crafted rules — learns its own feature embeddings.

So the question is: *at our dataset size, which approach works better?*

Experimental setup.



Stratified split

80 / 20

470 train · 118 held-out test



Cross-validation

3-fold

- Stratified 3-fold CV for classical ML
- Fixed train/val split for TabR



Hyperparameters

Optuna

TPE sampler
100 trials per model per fold



Primary metric

Balanced Accuracy.

+ F1, ROC-AUC, Precision, Recall

Test set never used during training or hyperparameter tuning · Framework: STREAMLINE + scikit-learn + XGBoost + Optuna

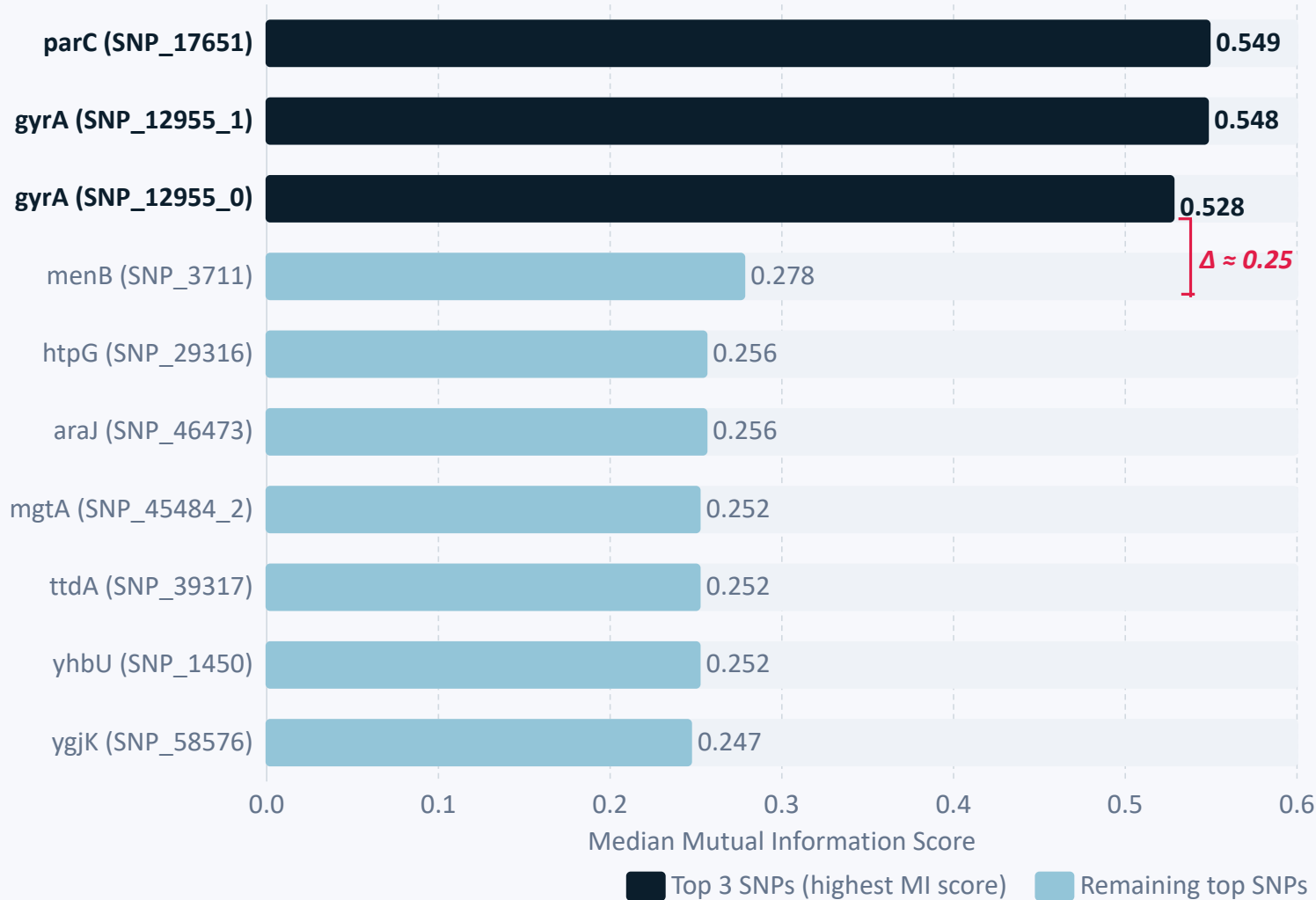
PART 02

Results

Feature importance, model performance, and the GPA vs SNP

Top SNP signals are driven by gyrA and parC mutations.

Top 10 SNP Features Ranked by Median Mutual Information Score



THE GAP

$\Delta \approx 0.25$

between rank 3 and rank 4

A small number of SNPs (gyrA, parC) dominate the signal—consistent with known ciprofloxacin resistance mechanisms.

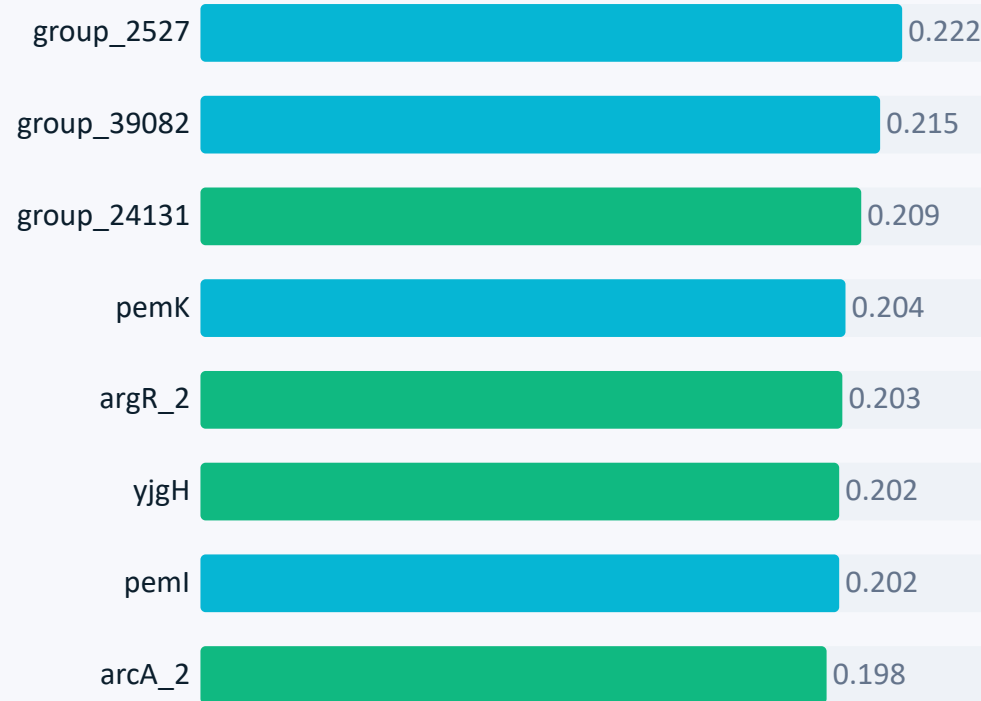
GPA: distributed signal across many genes.

Unlike SNPs, where three mutations dominate, GPA importance is spread across many genes.

✓ **13 of 20** top genes are shared across MI and MultiSURF -> consistent signal across methods

Mutual Information

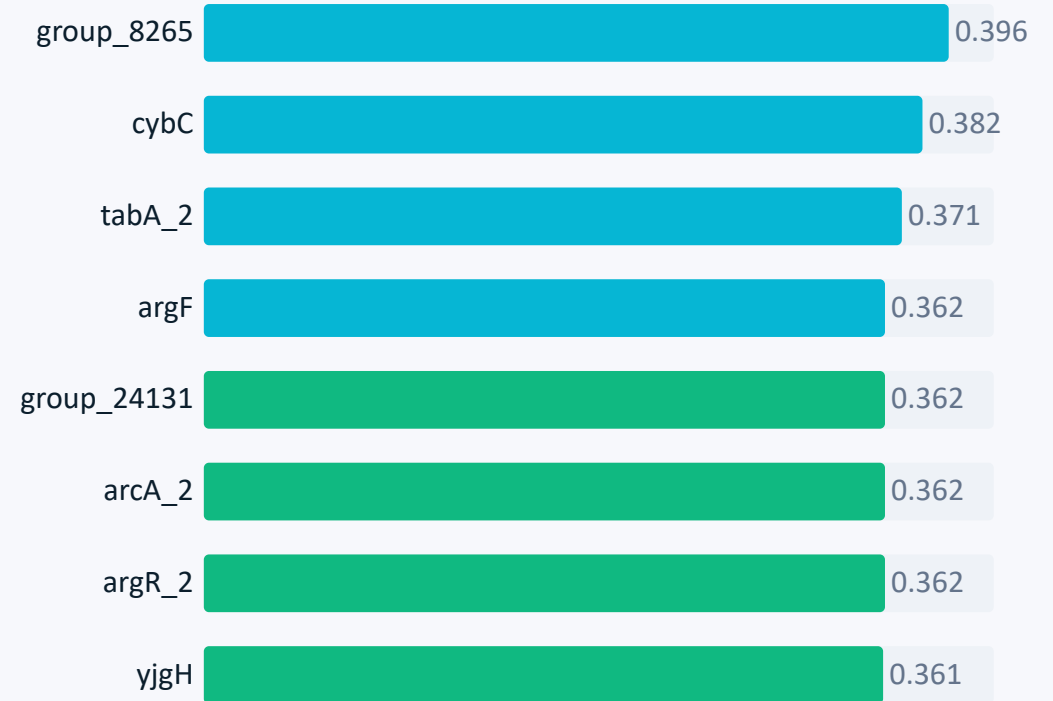
marginal feature relevance



■ Shared by both methods ■ Method-specific

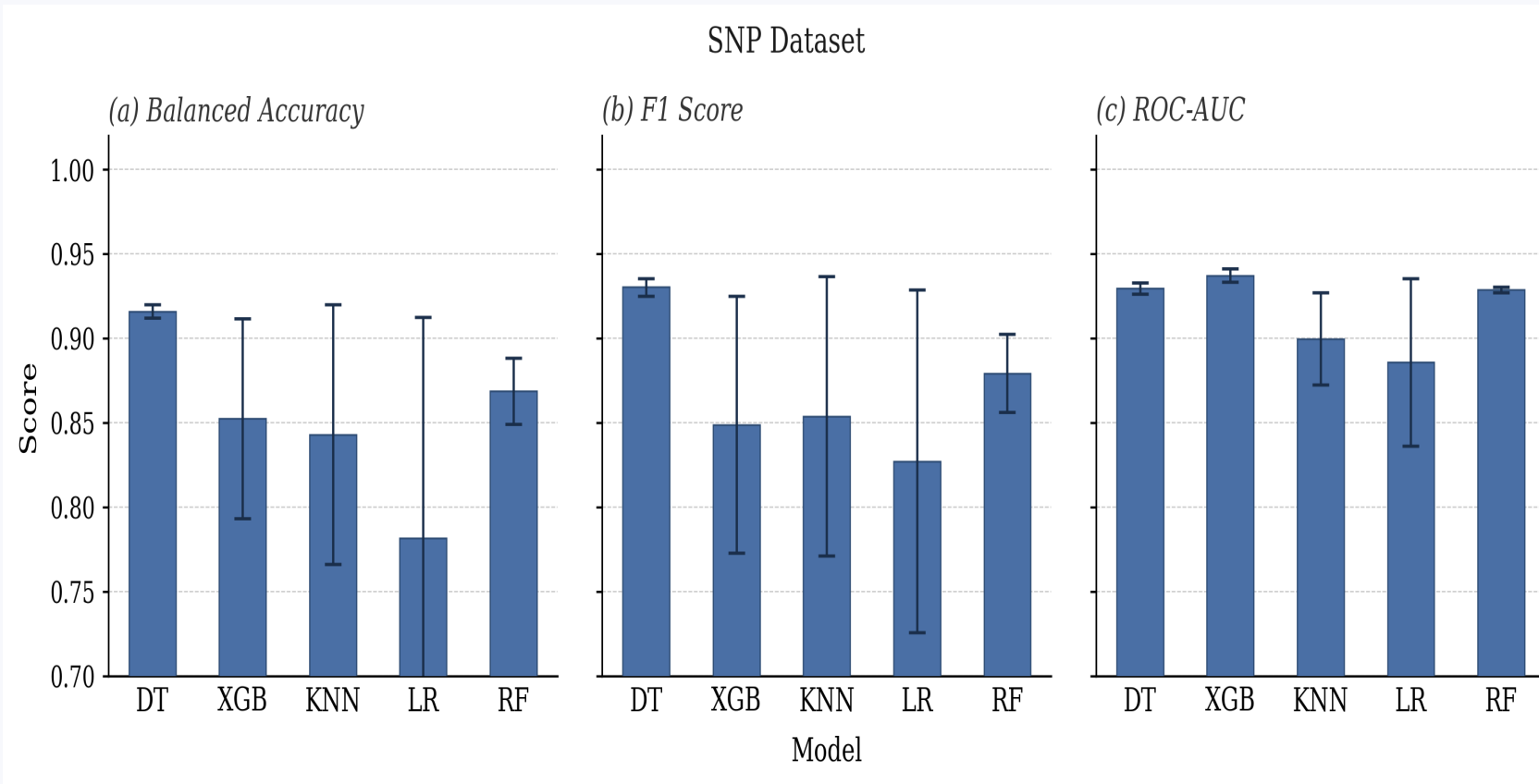
MultiSURF

neighbourhood + interaction signal



SNP test set : result

Primary metric: Balanced Accuracy — accounts for class imbalance in the resistance label.



Decision Tree — Best on SNP

Balanced Accuracy

0.916

F1 Score

0.930

ROC-AUC

0.929

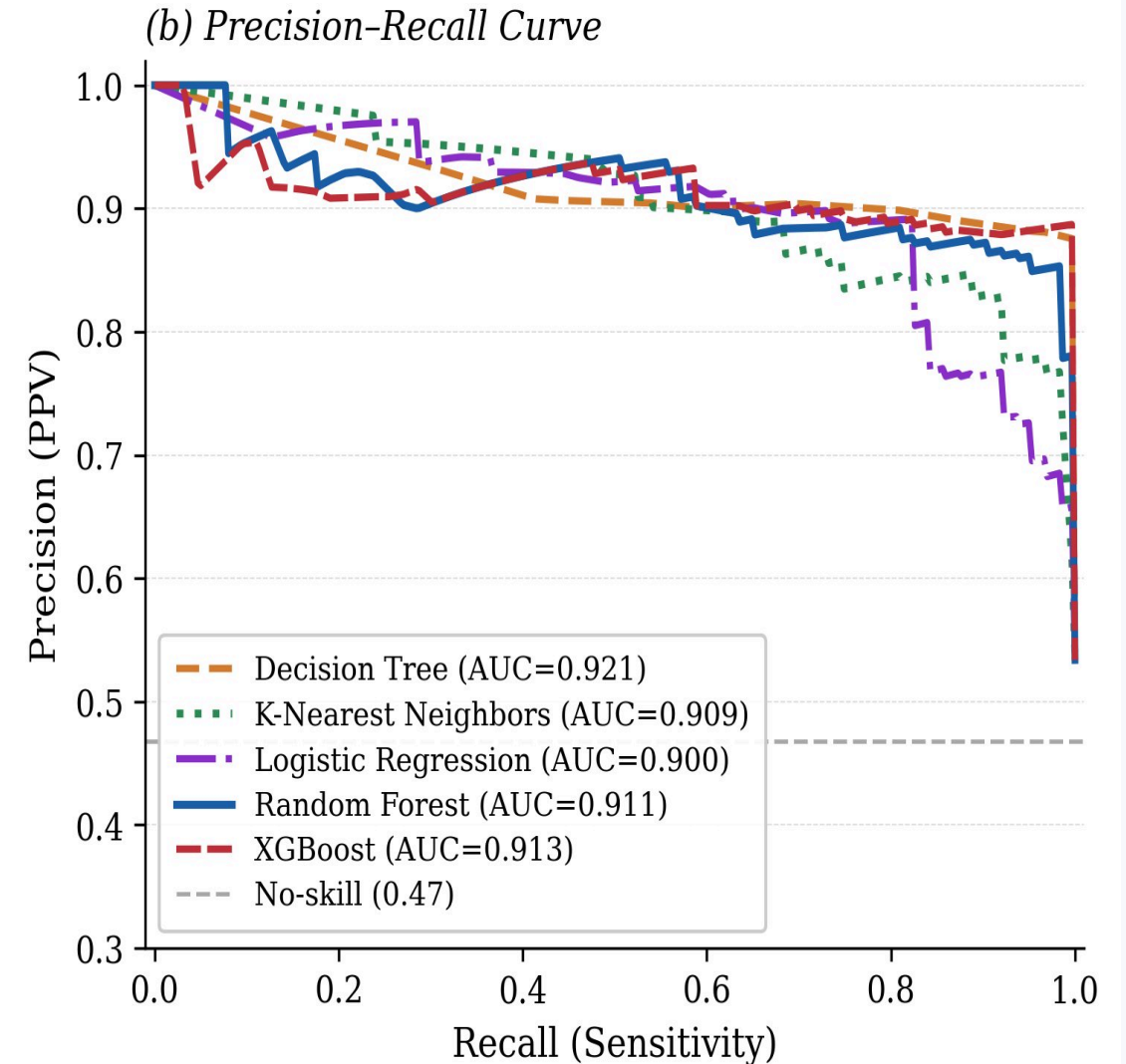
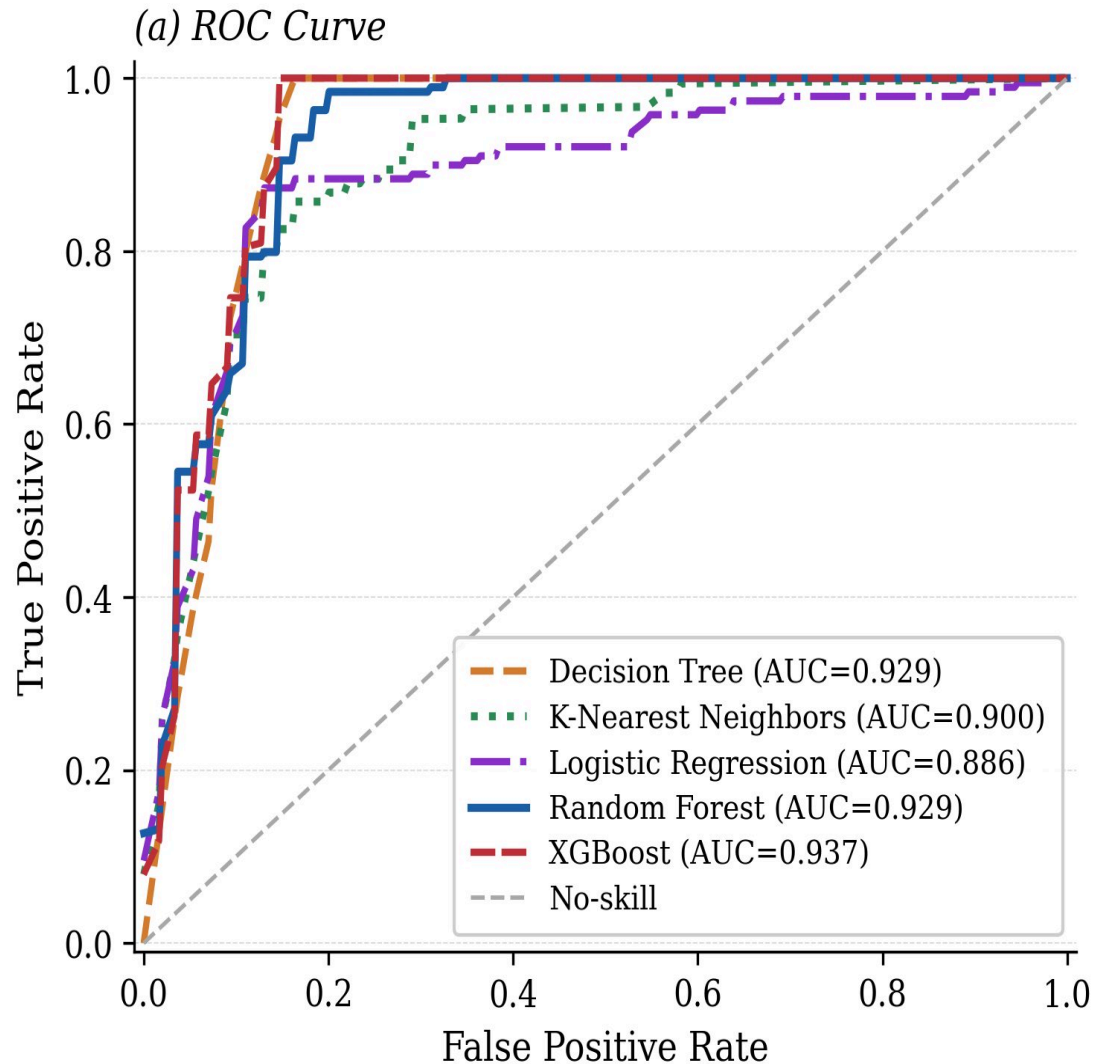
- ♦ XGBoost edges ROC-AUC at **0.937** — but trails on Bal. Acc.
- ♦ LR lowest (**0.782**) — linear boundaries fail in 47k-dim SNP space.

ROC & PR curves confirm the story — full curves in the next slide.

Tree-based non-linear structure captures *gyrA* / *parC* signal most effectively.

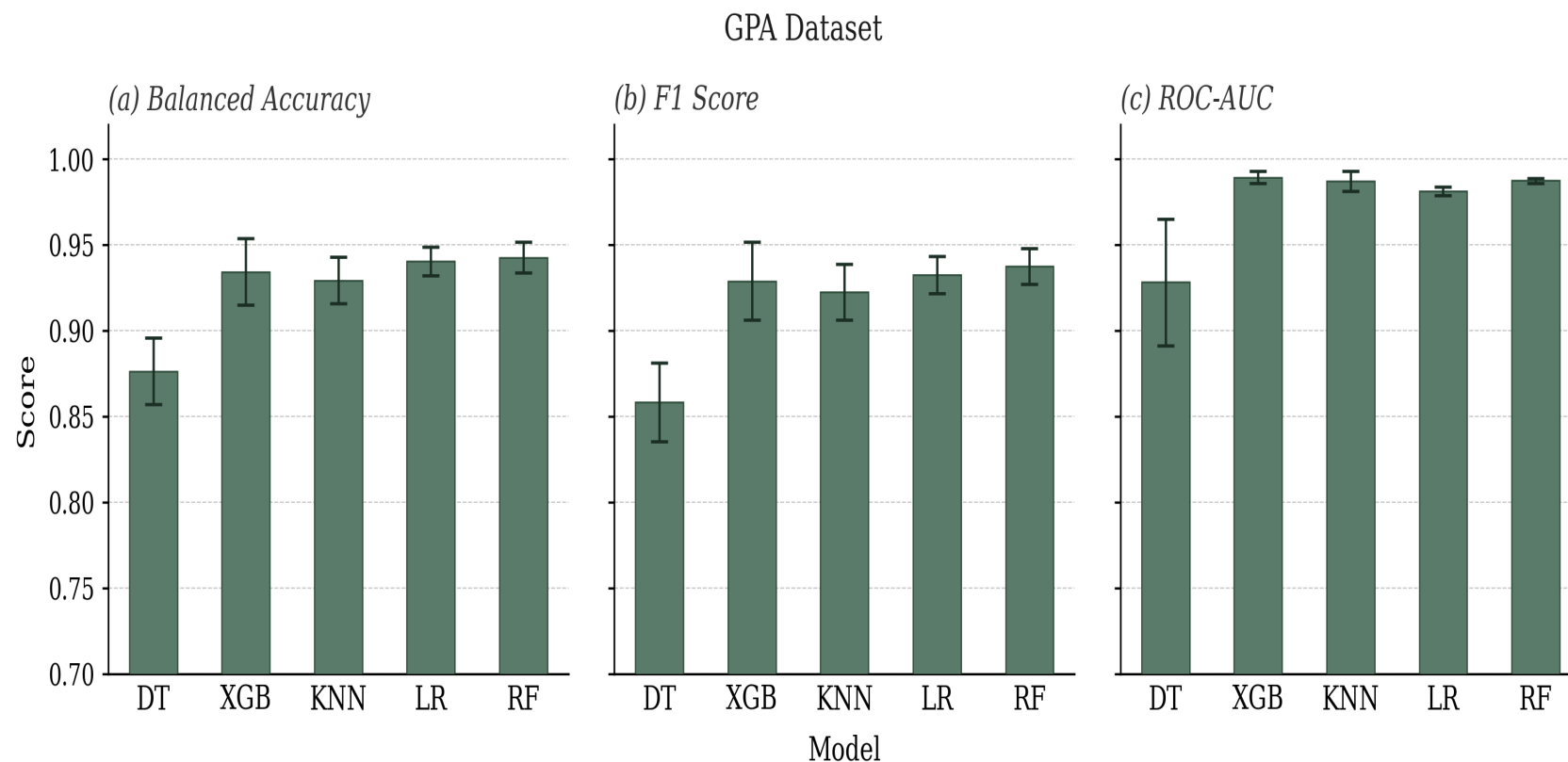
SNP: ROC and Precision-Recall curves.

XGBoost leads ROC-AUC (0.937) — all models well above no-skill baseline.



GPA test set: consistent performance across models.

Same metric, same structure — direct comparison with SNP results.



Random Forest — Best on GPA

Balanced Accuracy

0.943

F1 Score

0.938

ROC-AUC

0.987

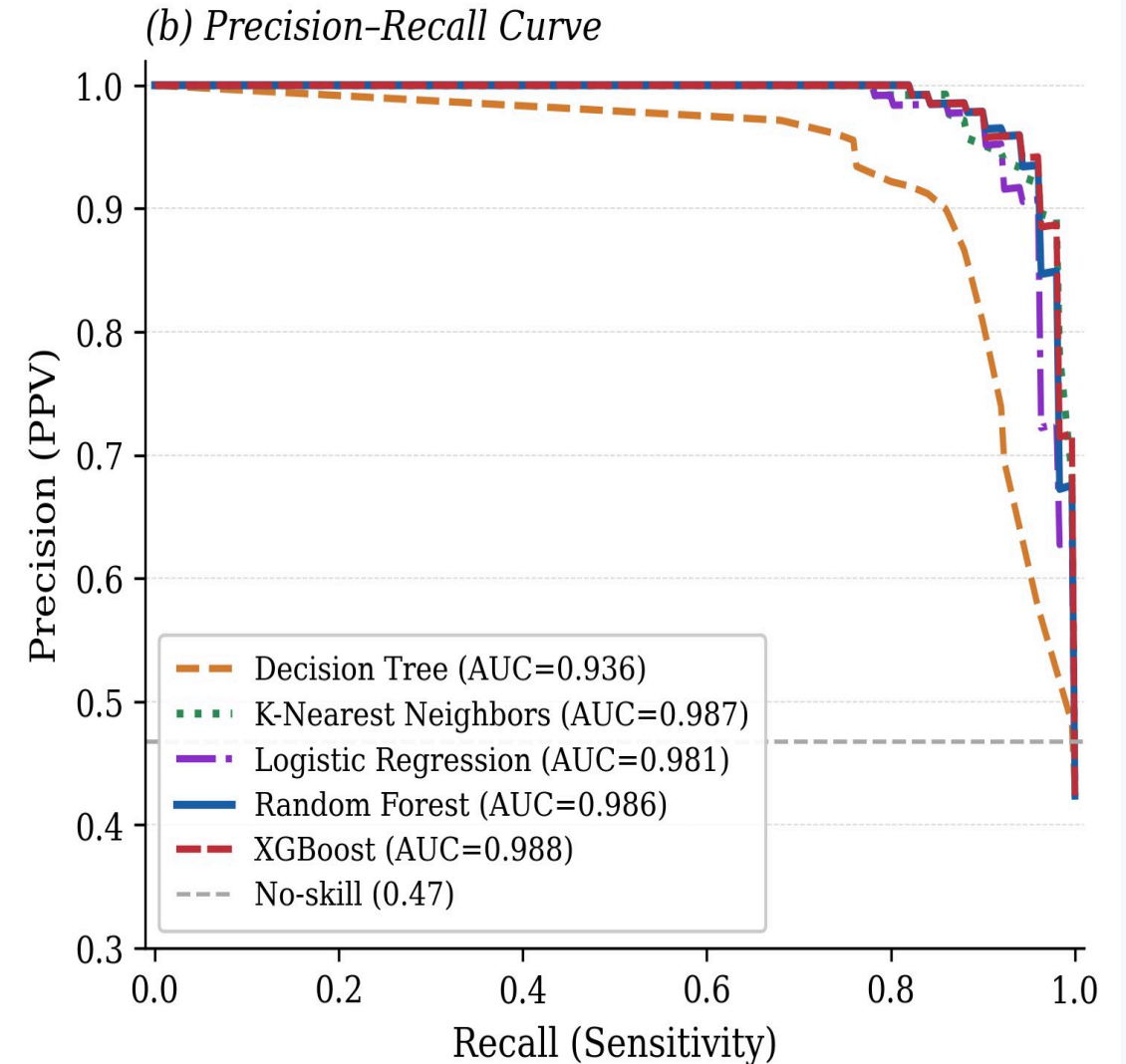
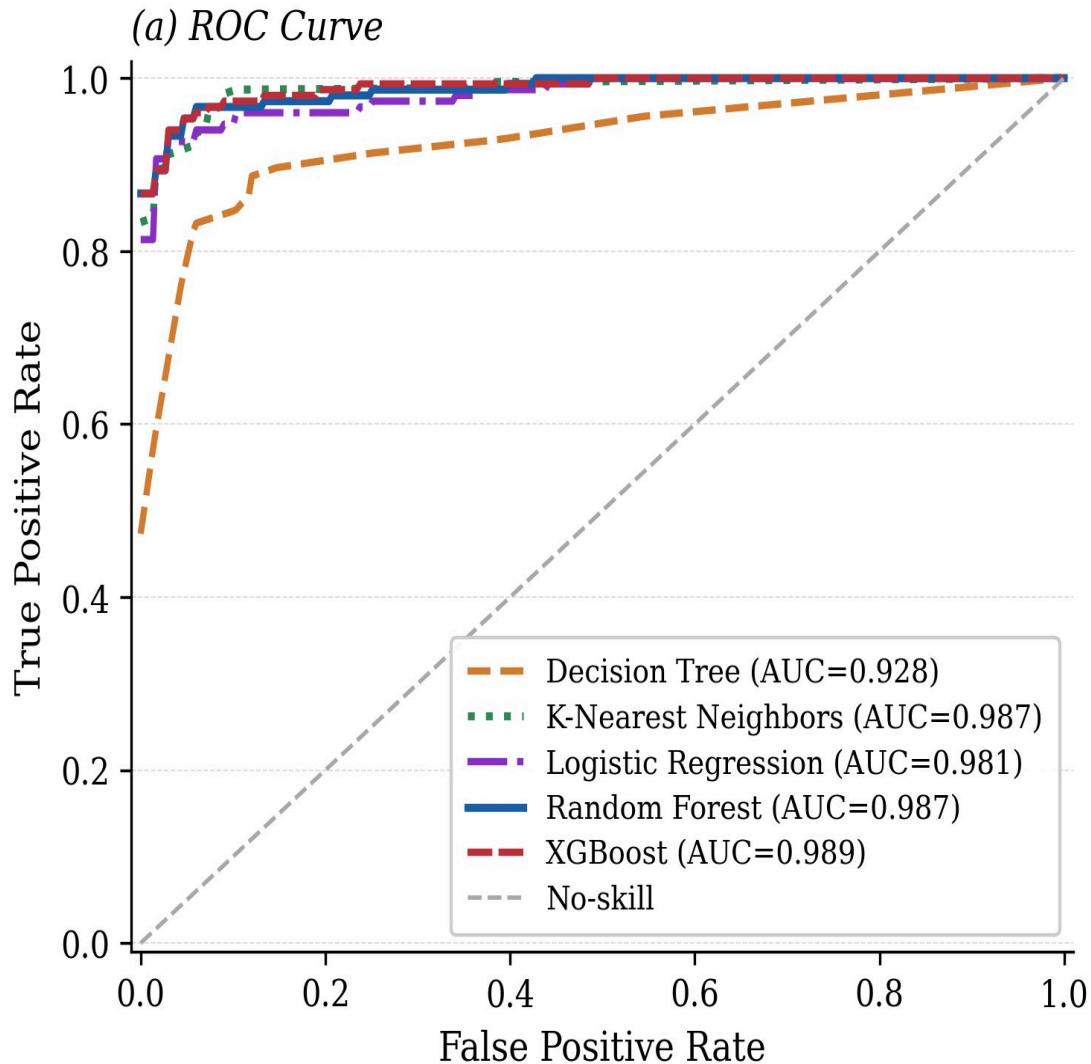
- ♦ Four models above **0.930** — far more consistent than SNP.
- ♦ Decision Tree lags (**0.876**) — shallow splits miss distributed gene signals.

ROC & PR curves confirm near-perfect discrimination — full curves in appendix.

Structured gene features reduce model sensitivity — GPA generalises better.

GPA: near-perfect ROC and Precision-Recall curves.

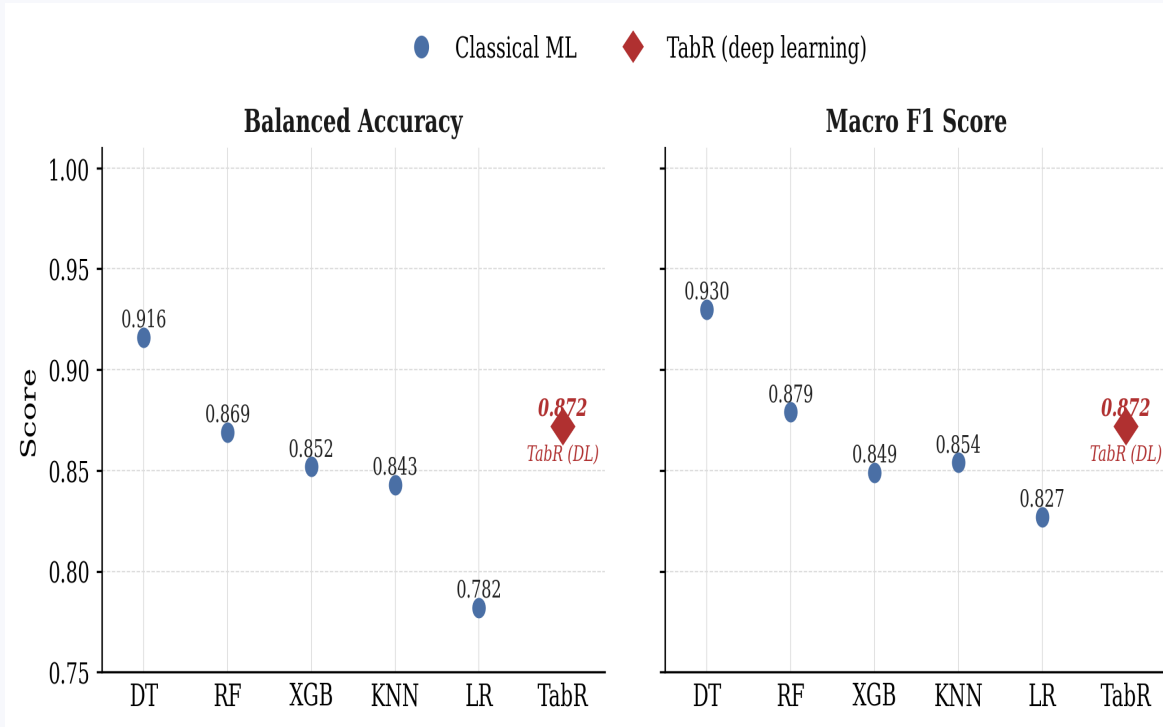
Four models above AUC 0.981 — GPA curves hug the top-left corner, far stronger than SNP.



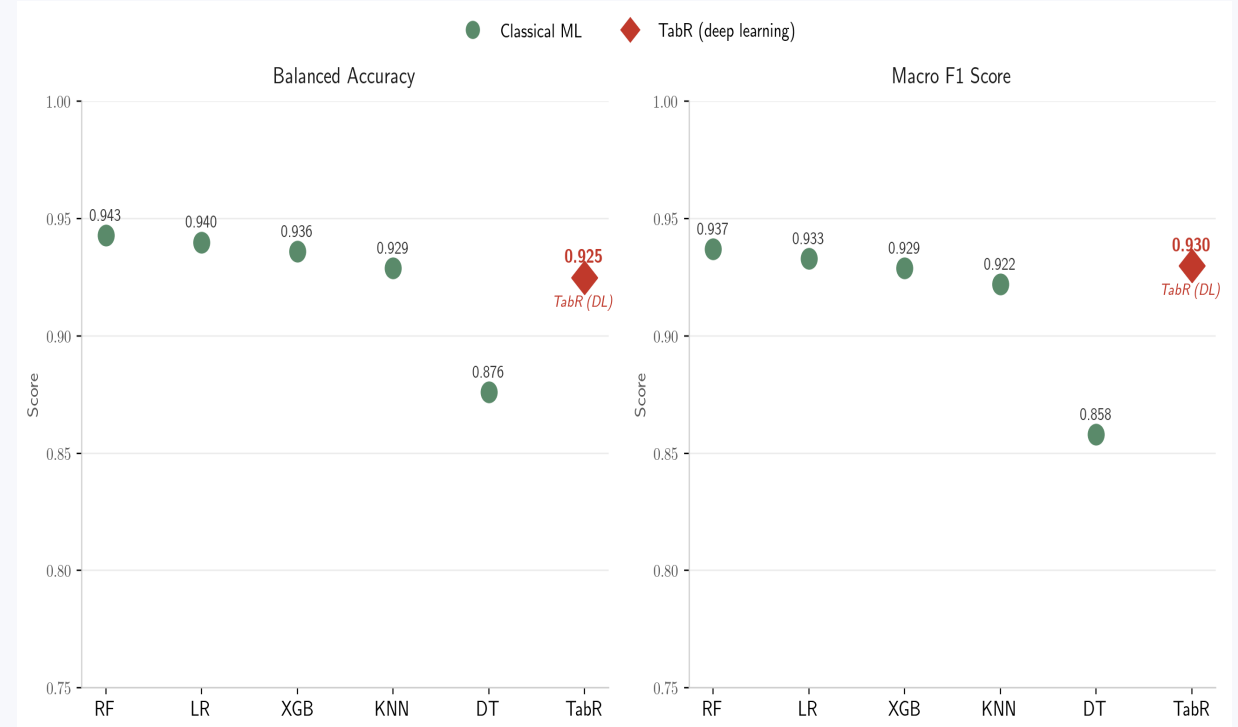
Classical ML outperforms TabR on both datasets.

TabR — retrieval-augmented deep learning — trails classical models on both SNP and GPA feature sets.

SNP Dataset



GPA Dataset



SNP

Best classical (DT) 0.916

TabR 0.872 $\Delta = -0.044$

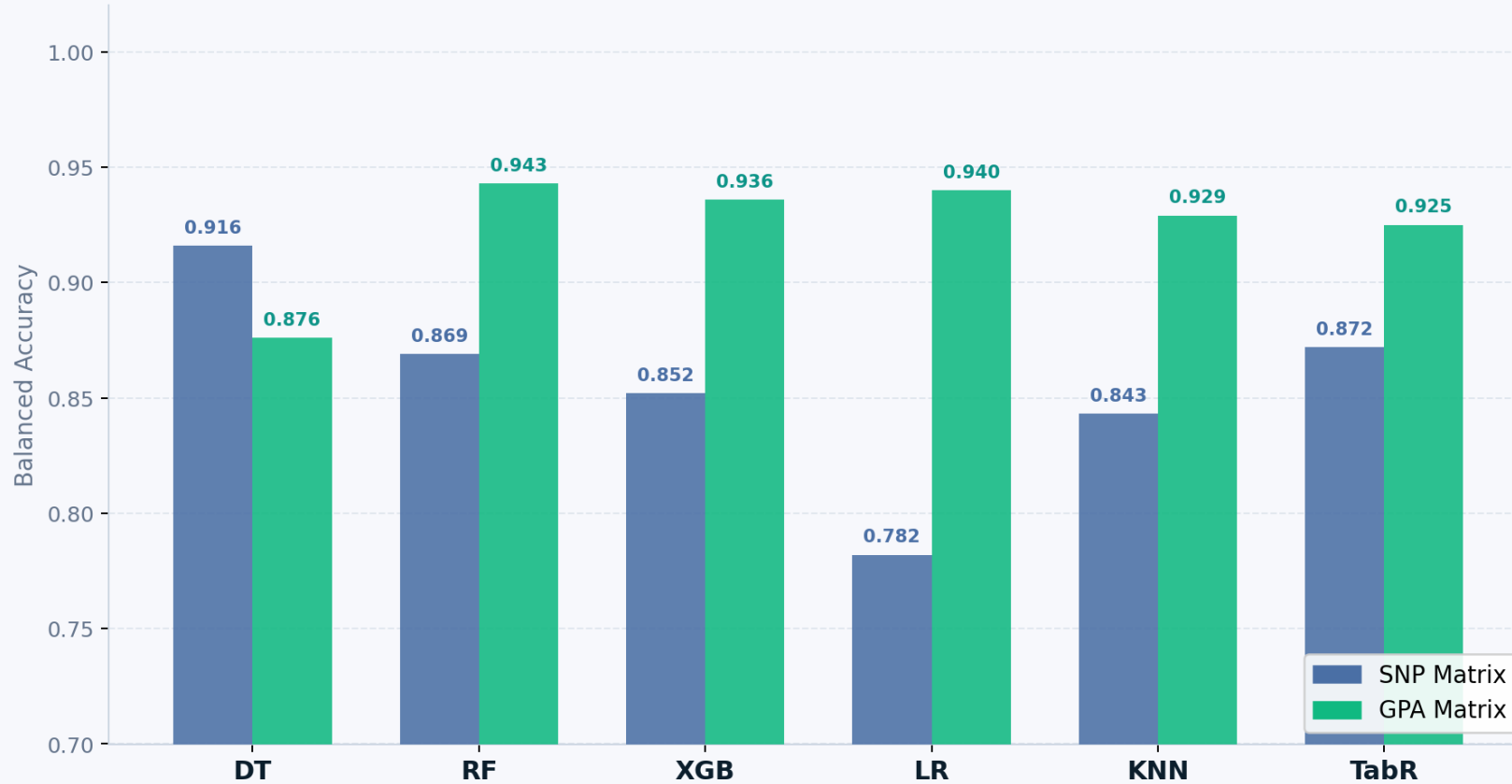
GPA

Best classical (RF) 0.943

TabR 0.925 $\Delta = -0.018$

TabR underperforms on both datasets — retrieval-augmented learning likely requires larger training sets than 588 isolates.

GPA outperforms SNP across every model except DT



✓ Same trend holds for F1 Score and ROC-AUC — GPA leads on both metrics across all models.

Best model per dataset

SNP · Decision Tree

0.916

GPA · Random Forest

0.943

Key takeaway

GPA outperforms SNP across all models

GPA gene presence/absence captures a cleaner, more structured signal for ciprofloxacin resistance.

Three findings, taken together.

01

GPA > SNP for prediction

★ Key finding

GPA features predict better than SNP features, even though SNP mutations in *gyrA*/*parC* are the known biological cause. Counter to our hypothesis

02

Classical ML > Deep Learning

Tree ensembles beat TabR on 588 isolates — deep learning needs more data.

03

Biology and ML agree on the signal

SNPs point to *gyrA*/*parC* mutations. GPA points to horizontally acquired resistance genes — ML captures both mechanisms.

These three findings together answer the central research question.

Honest about the boundaries.

Dataset size

588 isolates — sufficient for classical ML, insufficient for deep learning.

Shell genome only

Only shell genes used. Full species-wide pangenome not computed — some E. coli diversity may be missed.

No feature integration

SNP and GPA evaluated separately. Combined representations not explored.

Selection thresholds

Median MI threshold used for feature selection. MultiSURF not feasible at SNP dimensionality.

Despite these limitations, results are consistent and biologically meaningful.

What we found — and where to go next.

Conclusion

GPA features consistently outperform SNP features across all models — structured gene representations generalise better.

Classical ML (RF, LR, XGB) outperforms TabR on 588 isolates — deep learning requires more data.

gyrA and parC SNPs carry the dominant ciprofloxacin resistance signal — biology and ML agree.

A pangenome-based pipeline is a viable, scalable framework for AMR prediction in E. coli.

Future Work

- Integrate SNP + GPA into a unified feature representation to capture complementary biological signals.
- Apply transformer-based or graph neural network models designed for high-dimensional genomic data.
- Expand dataset with additional public E. coli bloodstream isolates to power deep learning.
- Extend framework to multi-drug resistance prediction beyond ciprofloxacin.

GPA features provide a more robust and learnable representation for predicting antibiotic resistance.

T H A N K Y O U

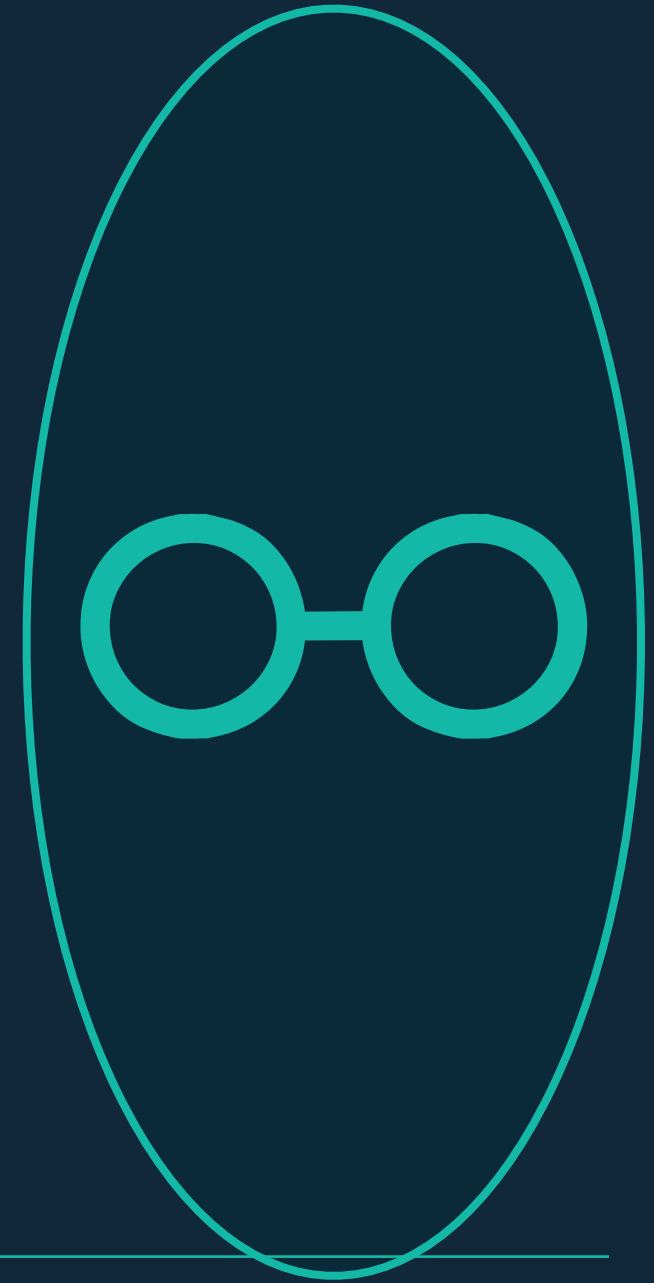
Questions?

A C K N O W L E D G M E N T S

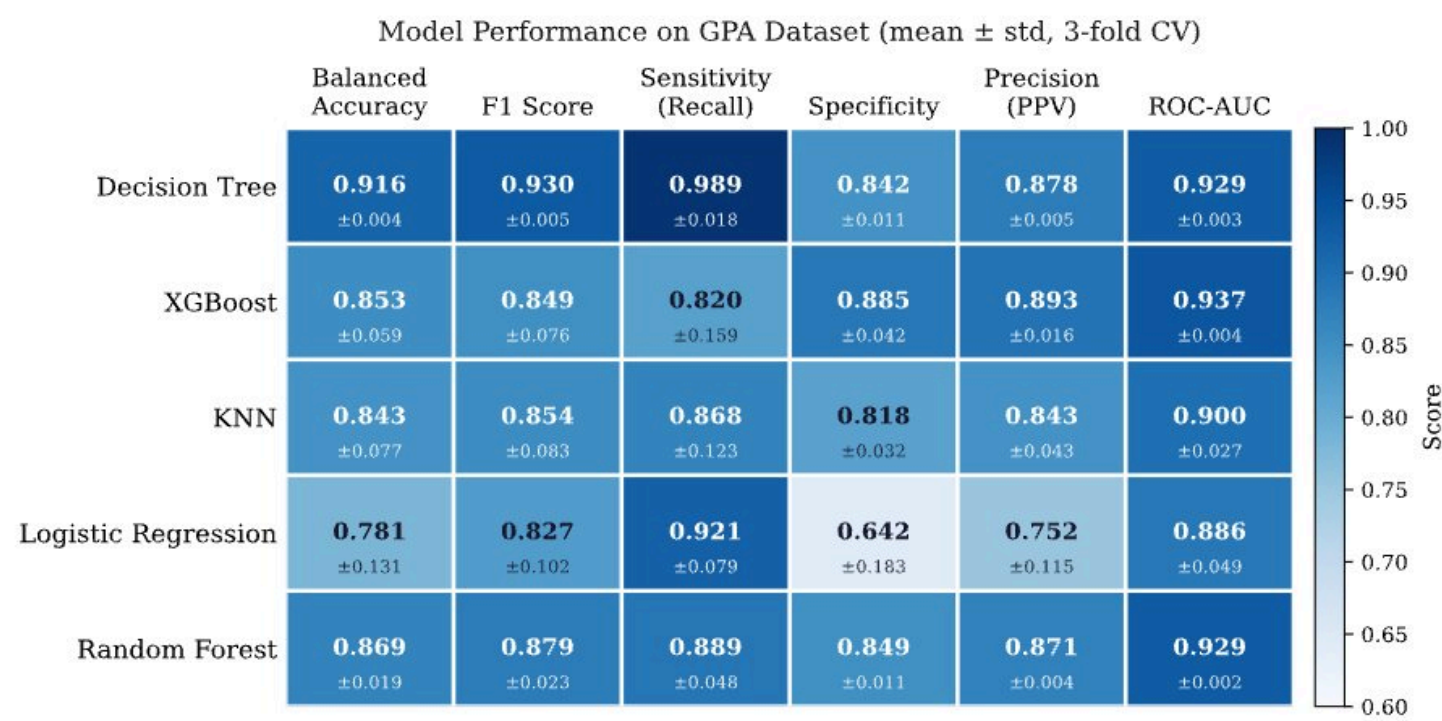
Committee: Dr. Sara El Alaoui · Dr. Akila de Silva · Dr. Pleuni S. Pennings

CodeLab team: Estefanos Kebebew, Myco Torest, Gian Carlo, Juve Barajas

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SNP Dataset — Model Performance



Mean ± std across 3-fold stratified CV. Note: in-figure title reads "GPA" — this panel reports the SNP dataset.

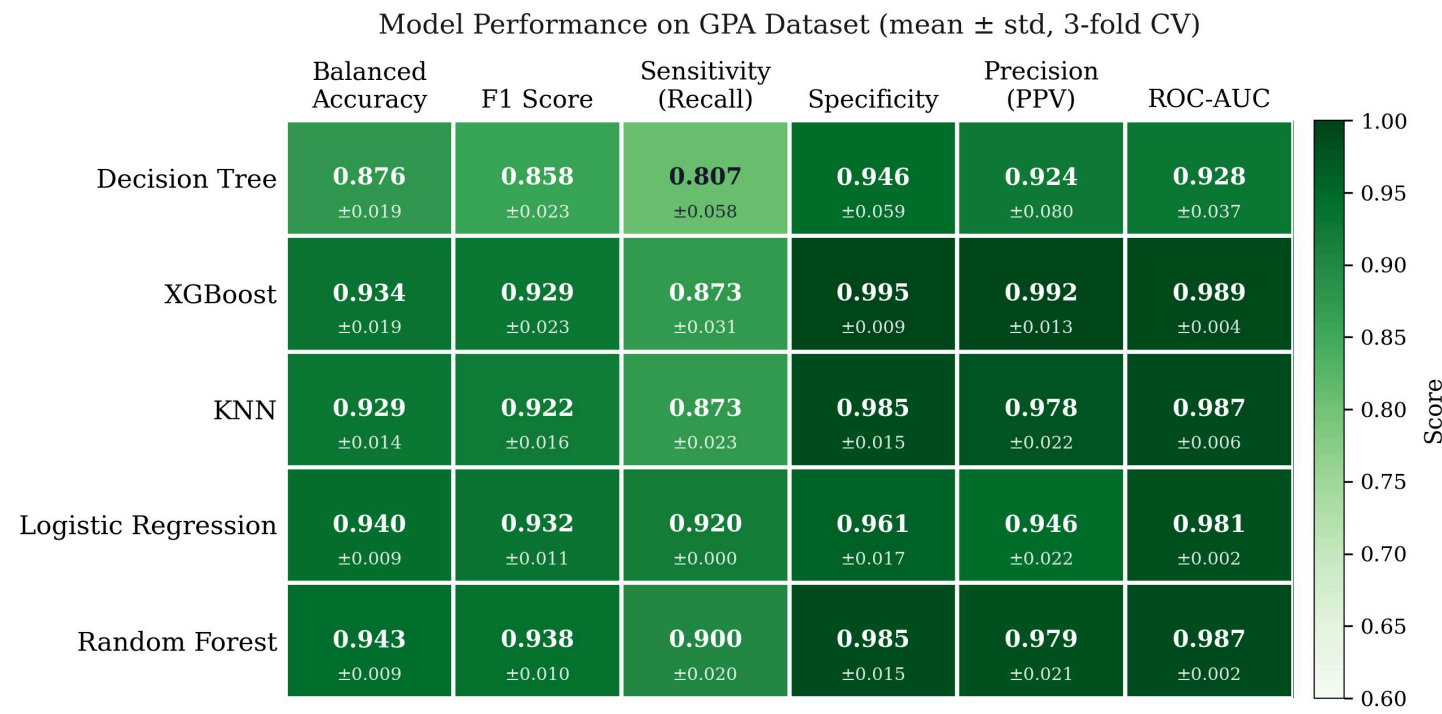
KEY OBSERVATIONS

Best: Decision Tree

Balanced Accuracy: 0.916 ± 0.004

- Decision Tree leads on BA, F1, Recall (0.989) — strongest sensitivity overall.
- XGBoost: highest ROC-AUC (0.937), best calibrated ranker.
- Logistic Regression: high recall (0.921) but low specificity (0.642) — high false-positive rate.
- Wide variance across models suggests SNP space is harder to learn than GPA.
- Feature space: 47,178 selected per fold.

GPA Dataset — Model Performance



Mean ± std across 3-fold stratified CV. Gene Presence/Absence binary feature matrix.

KEY OBSERVATIONS

Best: Random Forest

Balanced Accuracy: 0.943 ± 0.009

- All models ≥ 0.876 BA — GPA is consistently easier to learn than SNP.
- RF, LR, XGBoost cluster within 0.009 BA — robust signal across model families.
- XGBoost: highest specificity (0.995) and precision (0.992) — fewest false positives.
- ROC-AUC ≥ 0.981 across top 4 models — strong class separability.
- Feature space: 2,946 binary features after selection.

Selected Hyperparameters (Optuna, TPE)

100 Optuna trials per algorithm per fold · TPE sampler · objective: Balanced Accuracy · seed = 42

Classical ML — Best Hyperparameters

Model	Hyperparameter	SNP	GPA
DT	criterion	gini	gini / entropy
	max_depth	29	12 — 21
	min_samples_leaf	2 — 30	3 — 24
RF	n_estimators	381	354 — 496
	max_depth	18	23 — 27
	criterion	gini	gini / entropy
XGB	n_estimators	868	107 — 913
	max_depth	5	19 — 27
	eta (lr)	0.0072	0.0003 — 0.20
	subsample	0.510	0.524 — 0.776
KNN	n_neighbors	4 — 42	2 — 11
	weights	distance	distance / uniform
LR	solver	newton-cg / lbfgs / liblinear	newton-cg / sag / saga
	C	0.20 — 9.8e4	0.07 — 393.8

SNP folds (F0–F2) generally agreed on values; GPA shows wider per-fold variation. Range shown where folds differ.

TabR — Final Hyperparameters

Group	Parameter	Value
Training	batch_size	256
	n_epochs	100
	context_size	96
Model	d_main	256
	d_multiplier	2.0
	predictor_n_blocks	1
	context_dropout	0.2
Optim.	normalization	LayerNorm
	type	AdamW
	lr	3e−4
	weight_decay	1e−4

Identical hyperparameters across SNP and GPA datasets — suggests architectural robustness to feature representation.

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